

Powering The Future Data Centers: Murata's Role in Advancing OCP Innovation

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Strategic Power for AI

Power is no longer just an input. It's the critical enabler for **efficiency and AI innovation.**

- Murata is focused on power innovation to support exponential AI growth and system scalability
- Complete AC-to-xPU coverage: from 48/400/800 V distribution through intermediate conversion to >1000 A at the GPU rail
- Purpose-built platforms aligned with GPU and designed to support next-generation compute density, efficiency, and reliability
- Roadmap anchored in sustainability and TCO to meet the demands of future AI datacenters

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Active development for each power conversion stage.



400V/800V



50V



12V/6V



Power Cabinet
Sidecar

100kW AC to DC Shelf

HVDC

10kW & 6kW Bricks

IBC

Scalable Power - Bricks

VRM

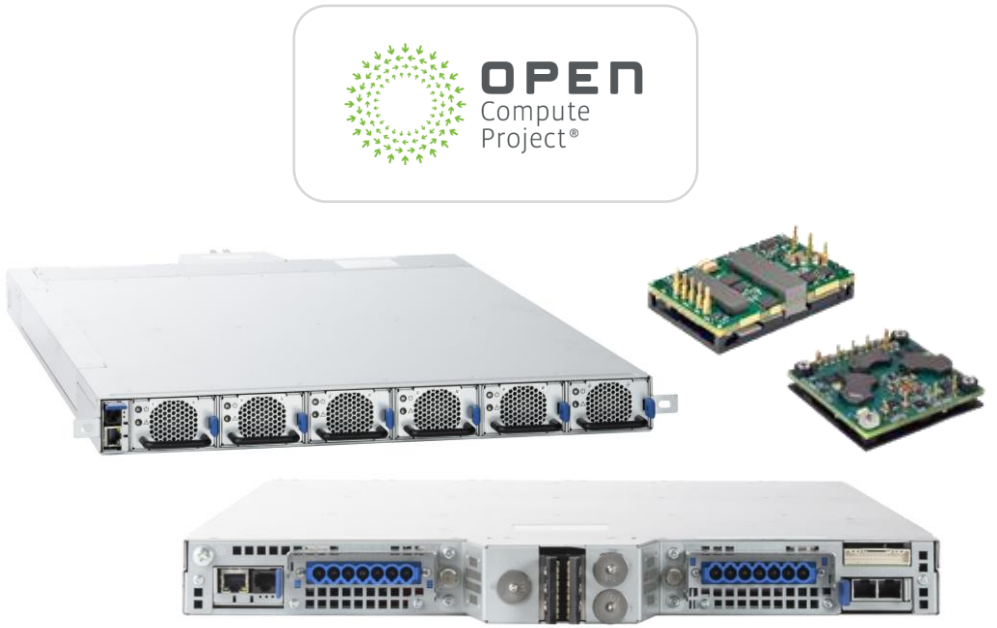
>1000A Vertical Power

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A Complete Power Ecosystem, Built to Scale

Murata is designing full-stack power systems **ready for the demands of today's AI infrastructure.**

- Bricks, converters, and shelves designed as a cohesive power ecosystem
- Designed for OCP compliance, thermal performance, and intelligent control
- Scalable platforms shipping today, with a roadmap tuned for higher density and efficiency



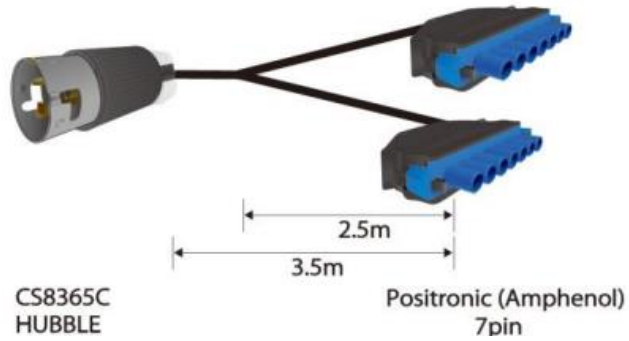
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Murata Supports ORV3 Powering Switches

- In the ORV3 system we created a PDU that fits in the back trough of the ORV3 rack. We use a Y cable to go from the 3-phase AC drop to this PDU and the shelf







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ORV3 21” or 19 “ Shelf

- Murata’s ORV3 is 21.6kW; 6 across 3.6kW Power Supplies
- Murata has both 50.5V and 54V version of the power supplies







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ORV3 HPR 33kW 21” or 19 “ Shelf

- Murata has both a 19” or 21” ORV3 HPR designed for OCP compliance and fit an MGX rack
- The 5.5kW power supply is 67mm wide to fit 6 across in either the 19” or 21” chassis. This provides a total of 33kW per shelf.
- Murata has a 50V and a 54.5V version module in development.



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Murata OCP Product Roadmap

OCP SPEC
3.0kW 50Vout PSU
18kW Shelf

OCP SPEC
5.5kW 50Vout PSU
33kW Shelf

OCP SPEC
12kW 50Vout PSU
72kW Shelf

Roadmap

50Vout Solution

50Vout, 800Vout Solution

Murata Roadmap

Shelf

PSU



Released
211-P-C
(21.6kW)
21”



Released
50V/3.6kW PSU
54V/3.6kW PSU



Released
191-P-C
(21.6kW)
19”

In Development
54V/5.5kW PSU (IQ4 26)



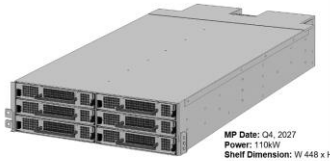
Released
211-P-D
(33kW).
21”



Released
191-P-D
(33kW).
19”



Released
50V/5.5kW PSU





In Development
54V/5.5kW PSU (IQ4 26)



Roadmap
(110kW)
19”



Roadmap
(72kW)
21”



Roadmap
50/54V/12kW PSU

23F 24F 25F 26F 27F

ORV3 Immersion Shelf









- 21” 3OU 54kW ORV3 Immersion Shelf as well as an 82.5kW ORV3 HPR Immersion Shelf
- The shelves use the ORV3 50V 3.6kW or ORV3 HPR 5.5kW power supplies modified to allow them to operate in dielectric immersion fluid.
- 50V and 54.5V versions are available.

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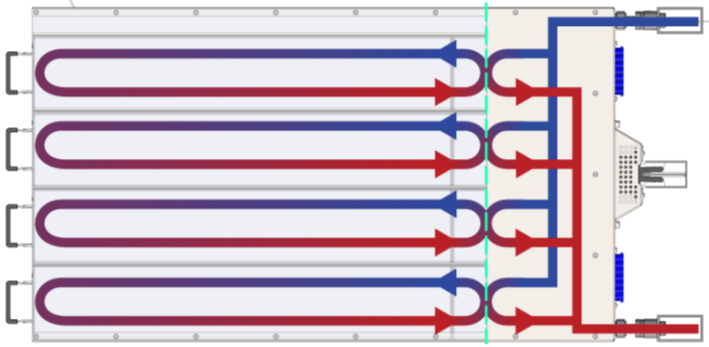
Designing to Scale for the Future and Beyond

Murata is pushing conventional topologies with next-gen architectures that **address density, cooling, and performance at scale.**

- Vertical power delivery architectures to support extreme density and layout flexibility
- Advanced cooling readiness including dielectric and direct-to-liquid solutions
- Modular HVDC platforms built for MW-class racks, hot-swap, and prefab deployment
- Power-aware system intelligence for rail-level telemetry and real-time fault management



Dielectric Cooling in development



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Not Just Power
– Murata's Comprehensive Component Solutions

- Murata is known for MLCCs – but our portfolio spans **much more.**
- Broad range of component technologies
  that complement system-level power architectures:
  - Capacitors & Inductors
  - Filters & Sensors
  - Thermal solutions etc.

- Together,
  these enable **today**'s AI systems and
  lay the foundation for
  next-generation AI infrastructure.



etc.

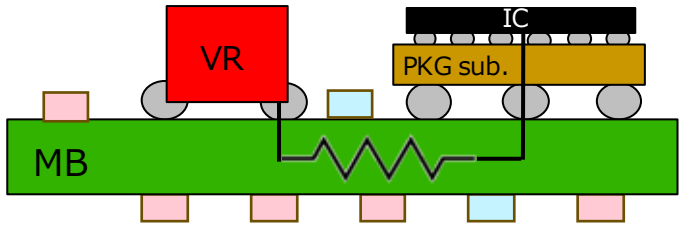
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Component Integration for the Chiplet Era

- Chiplet trend → multiple dies per package, each with unique power rails and high-speed I/O
- High power density + high transient loads → need for **near-load regulation** (VPD / iVR)
- Short power paths, low ESL/ESR, precise impedance control mandatory
- **Murata embedded component technology** enables:
  - Compact capacitors for on-package decoupling (MLCC, Silicon Capacitors)
  - High-current inductors for iVR designs
  - Thermal control with thermistors and vapor chamber wicks
  - Embedded capacitor/inductor solution for integrated, space-optimized module design

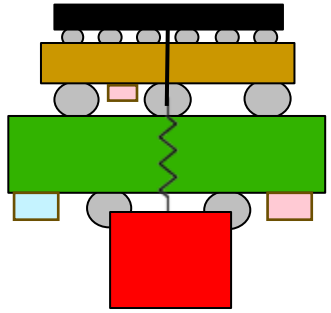
Traditional

Lateral Power Delivery



Next Step

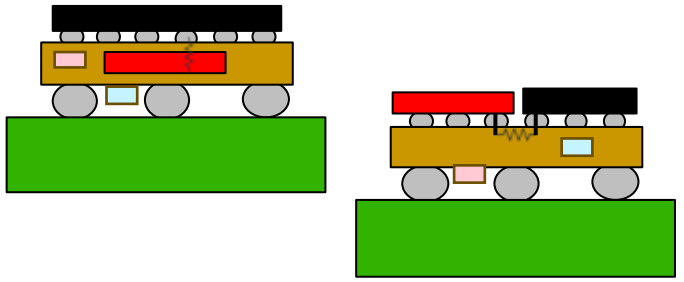
Vertical Power Delivery



Next Step

Future

On PKG VR（iVR）



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Murata's solution for advanced packaging & chiplet

Signal Integrity

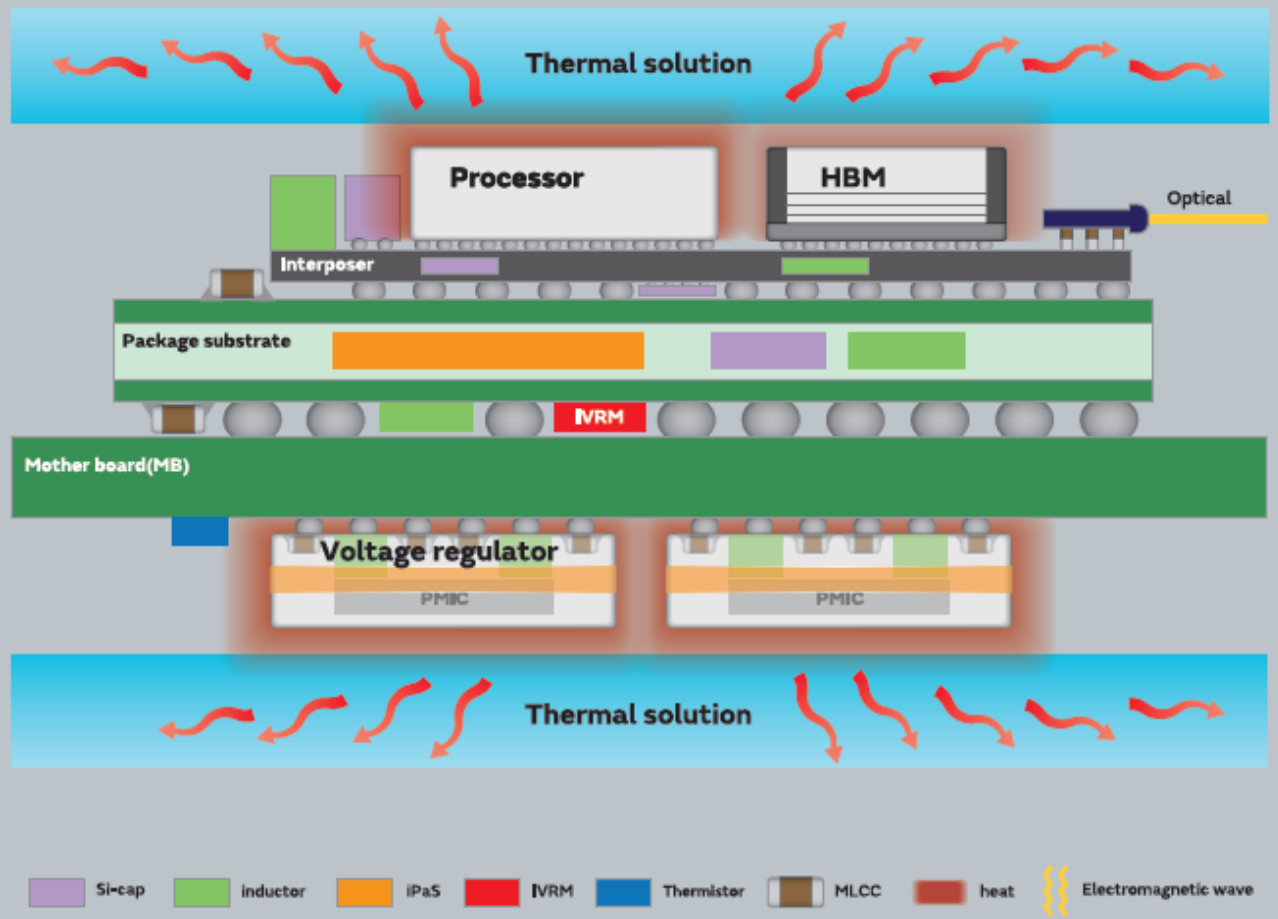
- MLCC
- Si-Cap
- Ferrite beads
- Common mode choke coil
- Timing Device
- Multilayer LCP product

Power Integrity

- MLCC
- Si-Cap
- Power Inductor
- iPaS(embedded capacitor/ inductor)
- IVR Module(R&D)
- DCDC Module
- Power Management IC

Thermal

- Vapor Chamber
- TIM (R&D)
- Thermistor



Chiplet-interconnects, RDL

- iPaS
- Multilayer LCP product

Photoelectric fusion

- Optical components

PDN Simulation Service

- Total PDN simulation (AC, Transient Analysis)
- Best suited, Best effort components recommendation

Design environment Improvement Service

- Murata Component Library integration into EDA tools
- Simulation model design/ customization (S-parameters, HFSS, etc)

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